

Om Goswami

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EDUCATION

Texas A&M University, Craig and Galen Brown Engineering Honors **May 2028**

B.S. Computer Science, Minors: AI in Business & Statistics – GPA: 3.93/4.0 — 4x Dean's List

- Clubs: Aggie Competitive Programming Club, TAMU Cybersecurity Club, Aggie Coding Club
- Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming, Software Engineering, Computer Systems, Functional Programming, Analysis of Algorithms, C++ Programming, Machine Learning and Predictive Modeling

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, JavaScript/TypeScript, SQL, HTML/CSS, Assembly, Haskell, R
- **Frameworks/Libraries:** FastAPI, Node.js, React, Streamlit, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, OpenCV, NLTK, LiteLLM
- **AI/ML & MLOps:** LLMs, RAG, MCP, Agentic Workflows, Prompt Engineering, NLP, YOLOv8, ChromaDB, MLflow
- **Databases & Cloud:** AWS (S3, Glue, Lake Formation, Athena, DataZone), Databricks (Delta Lake, Unity Catalog), Firebase, PostgreSQL, MySQL, NoSQL
- **Developer Tools:** Git, GitHub, Claude Code, Unix/Linux, Docker, REST APIs, VSCode, Jupyter
- **Certifications:** NVIDIA – Building RAG Agents with LLMs, NVIDIA – LLM Applications with Prompt Engineering, Anthropic – AI Fluency, Visa – Payments Cybersecurity

EXPERIENCE

Visa | Software Engineering Intern | *Python, React, JS, FastAPI, PostgreSQL, LiteLLM, Docker* **May 2026 – Present**

- Design, test, and deploy the backend for an internal **Vendor Intelligence Portal** in an **Agile** environment, deploying through a **CI/CD** pipeline to a live production environment used by **50+ analysts and engineers** across multiple Visa teams, using **FastAPI, PostgreSQL, and Pydantic** to build scalable **REST** APIs, with contributions to the **React** frontend and **QA** testing.
- Develop automation **scripts** orchestrating **Claude Sonnet** and **Opus** via **LiteLLM** to automate vendor intelligence extraction, document parsing, and generation of PxQ tables and TARs, reducing manual research and report generation from hours to minutes.
- Analyze and deliver high-volume automated **agentic workflows** to reduce manual analysis and reporting effort by **~70%**, lower operational costs by an estimated **~30%**, and accelerate procurement decision-making by eliminating repetitive manual processes and saving hundreds of analyst hours annually.

Texas A&M University | Undergraduate Teaching Assistant **Aug 2025 – Present**

- Teach **Python** fundamentals – loops, conditionals, functions, and data structures – to **200+** first-year engineering students each semester through weekly office hours, **1:1 mentoring**, and hands-on debugging sessions.
- Perform **code review** on programming assignments and deliver detailed, code-level feedback on software design, code quality, and debugging practices, strengthening student fundamentals and technical communication.

TRAIN | Artificial Intelligence Fellowship **Jan – Apr 2024**

- Designed and built an end-to-end AI-powered educational tool to simplify machine learning concepts for beginners, implementing core algorithms in **Python** and training **NLP** models with the **Transformers** library.
- Integrated the **Hugging Face API** to power a conversational **Q&A** interface, letting beginners ask natural-language questions about ML concepts and receive real-time, contextual explanations.

PROJECTS

AI-Powered Incident Response & SRE Platform | *Python, FastAPI, React, LiteLLM, ChromaDB* | [Github](#) **Jul 2026**

- Built an autonomous AI platform that continuously monitored a production-style web service under **~450 requests/sec**, automatically detecting software failures, performing **root cause analysis** to identify faulty **Git** commits, and coordinating AI-assisted incident response.
- Developed an **LLM**-powered investigation pipeline that analyzed **Git** commit history and code diffs, retrieved operational runbooks using **RAG**, estimated business impact, and generated Slack incident reports for engineers.
- Implemented human-in-the-loop remediation by automating **Git** rollback, redeployment, and live metric verification, validating recovery across **4** realistic production failure scenarios.

Smart Traffic Monitoring and Alert System | *React, Vite, Tailwind, Flask, Twilio* | [Demo](#) | [Github](#) **Jan 2026**

- Built **Oculon**, a dashboard aggregating **12 live HLS** feeds from **4** state DOT camera networks that flags collisions by matching playback position to incident windows derived from **frame-level motion analysis**.
- Engineered a pluggable alert pipeline dispatching incident reports (type, location, severity, time) to **Discord** with fallback to **Twilio SMS**, and fixed cross-browser playback via **hls.js**.
- Corrected a mistimed collision window via frame-by-frame analysis, replaced dead camera feeds, and shipped the **React/Flask** app as one **Vercel** serverless deployment.

End-to-End Data Engineering & MLOps Pipeline | *AWS, Databricks, PySpark, PyTorch, MLflow* **Jun 2025 – Jul 2026**

- Designed a governed data lake on **AWS S3** with **Glue** for ETL/cataloging and **Lake Formation** for access control, publishing datasets through **DataZone** for discovery.
- Extended the pipeline into **Databricks**, building a **distributed**, bronze-silver-gold **Delta Lake** pipeline in **PySpark** to clean and reshape raw Census data into versioned tables in **Unity Catalog**.
- Trained a **PyTorch** neural network to predict state-level population growth, tracking experiments and registering model versions through **MLflow** for **MLOps** traceability.

LEADERSHIP EXPERIENCE

BAPS | Project Manager, Regional Core Team Lead | *Collaboration, Communication, Problem-Solving* **Aug 2020 – Present**

- Orchestrate a large-scale event for **600+ delegates** from the southwest region, featuring skits, light shows, and presentations.
- Develop and implement national guidelines for **2500 leads**, contributing to a **35%** increase in attendance over the past year.

Perplexity AI | Marketing Strategist **Jan – May 2025**

- Led growth initiatives at Texas A&M, introducing **750+ students** to Perplexity's AI while testing **5+** acquisition strategies with Perplexity's Growth Team – driving a **35%** increase in weekly engagement and **600+** student sign-ups.